



## **Digital Exclusion of Traditional Textile Weavers in Rural India**

Research Paper

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### Abstract

This study examines the digital exclusion of traditional textile weavers in rural India, analyzing its impact on market competitiveness and economic sustainability. Despite the rapid digitization of global markets, over 70% of rural weavers lack digital literacy, and nearly 60% have no access to reliable internet, limiting their participation in e-commerce and direct sales (Ansari & Tripathi, 2018; Singh et al., 2018). The heavy reliance on intermediaries results in lower earnings, with weavers losing up to 40% of their potential revenue due to exploitative supply chains (Ghosal et al., 2020; Meher et al., 2024). Using a secondary data analysis approach, this research synthesizes findings from recent studies and industry reports to evaluate the extent of digital adoption among weavers and its economic implications (Briggs-Goode & Townsend, 2016; Kumar et al., 2022). The findings indicate that digital illiteracy, lack of infrastructure, and inadequate policy support significantly hinder weavers' ability to access digital platforms, further exacerbating economic marginalization (Sangwan & Kumar, 2024; Digital Empowerment Foundation, 2021). To address these challenges, this paper proposes a **Community-Centric Digital Empowerment Model**, focusing on digital literacy training, infrastructure investment, direct-to-market strategies, and sustainable business integration (Brynjolfsson & McAfee, 2014; Priya & Shrivastava, 2020). Policy recommendations emphasize the urgent need for targeted digital inclusion initiatives, government support programs, and improved financial accessibility to ensure the long-term economic resilience of traditional artisans in the digital era (Carpenter & Poon, 2023; Ibragimova, 2019).

*Keywords:* digital exclusion, rural weavers, e-commerce barriers, digital literacy, digital empowerment, market access, rural India

## Introduction

### Unveiling Digital Exclusion: The Plight of Traditional Textile Weavers in Rural India

The rapid digitalization of global markets has transformed industries, yet India's traditional handloom sector, one of the largest employers in rural areas, remains largely disconnected from this shift (Meher et al., 2024; Digital Empowerment Foundation, 2021). The handloom industry is not only an essential economic contributor, employing over 4.3 million artisans across 2.4 million households, but also a vital custodian of India's rich cultural heritage, producing globally recognized textiles that reflect centuries-old craftsmanship (Mukhopadhyay, 2022; Niranjana, 2001). Despite its significance, the sector faces a persistent challenge: **digital exclusion**, which limits market access, reduces economic sustainability, and reinforces dependence on traditional, often exploitative, supply chains (Kanupriya, 2022; Basole, 2016).

Digital illiteracy, inadequate infrastructure, and socio-economic barriers prevent rural weavers from leveraging digital platforms for business expansion, placing them at a competitive disadvantage in an increasingly digital economy (Ansari & Tripathi, 2018; Kumar et al., 2022). Studies indicate that over 70% of handloom weavers lack digital literacy, while nearly 60% have no access to stable internet, significantly restricting their ability to engage in e-commerce and direct-to-consumer sales (Singh et al., 2018; Ghosal et al., 2020). The reliance on intermediaries leads to revenue losses of up to 40%, further undermining economic sustainability (Sangwan & Kumar, 2024; Pande, 2022).

This research investigates the economic and social implications of digital exclusion among rural weavers, highlighting key structural barriers and proposing a **Community-Centric Digital Empowerment Model** to bridge the digital divide (Briggs-Goode & Townsend, 2016; Ibragimova, 2019). By integrating digital literacy training, infrastructure development,

and direct-to-market strategies, this model aims to enhance weavers' access to the digital economy while preserving their cultural legacy (Brynjolfsson & McAfee, 2014; Priya & Shrivastava, 2020).

## **Literature Review**

### **Digital Illiteracy**

Digital illiteracy remains a primary obstacle for traditional weavers in rural India, hindering their ability to engage in modern markets (Ansari & Tripathi, 2018). A significant majority of weavers—over 70%—lack the necessary digital skills to navigate e-commerce platforms, conduct digital transactions, or employ online marketing strategies (Bach et al., 2013). This digital divide not only restricts market access but also limits the weavers' ability to expand their business beyond local markets (Singh et al., 2015). While some training programs exist, they are primarily focused on urban entrepreneurs, leaving rural artisans underprepared (Anrijs et al., 2023; Singh et al., 2015; Liu & Liu, 2023). Additionally, most rural artisans remain reliant on traditional sales channels, including intermediaries, who reduce their profit margins (Mukhopadhyay, 2022). According to Brynjolfsson and McAfee (2014), digital empowerment can help break this cycle by providing weavers with the knowledge and tools needed to leverage digital platforms, which is essential for economic growth in the sector.

### **Infrastructure Gap**

The infrastructure gap remains a significant barrier to the digital inclusion of rural weavers. Poor internet connectivity, unreliable electricity, and a lack of affordable digital devices hinder their access to digital platforms (Potnis, 2016). Approximately 60% of weavers in rural India lack consistent internet access, and almost 50% do not own smartphones or computers (Singh et al., 2018). This technological deficit severely limits the

weavers' ability to engage in e-commerce or access broader, global markets (Brynjolfsson & McAfee, 2014). Singh et al. (2015) argue that without addressing the basic infrastructural challenges, efforts to digitally empower rural artisans will likely fail. Furthermore, despite efforts by government initiatives and NGOs, infrastructure-related obstacles persist, leaving the weavers at a significant disadvantage compared to urban counterparts (Digital Empowerment Foundation, 2021). These infrastructural limitations compound the digital illiteracy issue, making it difficult for weavers to adopt and sustain digital technologies effectively (Singh et al., 2018).

### **Critical Analysis and Limitations**

Although digital inclusion efforts have been made through government programs and NGOs, these initiatives have not yielded substantial or long-term impacts for rural weavers. Existing programs often fail to provide adequate financial support, preventing weavers from purchasing necessary digital tools (Singh et al., 2018). In addition, many training programs do not address the specific needs of rural artisans, leading to limited understanding of how digital tools can help improve their business (Casciani et al., 2022). While some research has explored the impact of digital exclusion on rural industries, there is a lack of empirical studies that directly examine the challenges faced by traditional textile weavers (Helsper, 2012; Bach et al., 2013). This gap in literature calls for a more comprehensive approach to integrating digital tools and community-driven empowerment strategies to support rural weavers (World Bank, 2020; Priya & Shrivastava, 2020).

### **Cultural and Economic Importance of the Handloom Sector in India**

The handloom industry holds significant cultural and economic importance in India, contributing to both the nation's heritage and its economy. Traditional textiles such as Banarasi, Chanderi, and Kanjeevaram hold cultural value and support regional identity

(Singh et al., 2015). The sector generates \$3.5 billion annually and accounts for nearly 15% of India's textile production (Pande, 2022). However, the rise of automation and the digital divide has led to a 30% decline in handloom employment over the past two decades (Ghosal et al., 2020). This decline is part of a broader trend where traditional artisans struggle to compete with industrialized textile production that benefits from digitalization (Anrijs et al., 2023). While some argue that digital tools have enhanced industrial textile production, they have marginalized rural weavers who lack the infrastructure and skills necessary to thrive in this digital age (Singh et al., 2018). As a result, these artisans continue to face challenges in accessing global markets and gaining financial independence through digital means (Niranjana, 2001).

## **Methodology**

### **Research Approach & Rationale**

This paper adopts a **secondary data analysis** approach, focusing on the synthesis of existing literature, government reports, and case studies related to digital exclusion in the handloom sector. The research is primarily **conceptual** in nature, drawing insights from previous studies, surveys, and interviews conducted by **organizations like the Digital Empowerment Foundation and the World Bank** (World Bank, 2020; Digital Empowerment Foundation, 2021). This approach is appropriate as it enables a comprehensive understanding of the barriers to digital adoption faced by rural weavers, allowing for a synthesis of key trends and identifying potential solutions (Sangwan & Kumar, 2024). By analyzing the available secondary data, this study aims to uncover the underlying issues and opportunities for addressing digital exclusion in the handloom industry (Priya & Shrivastava, 2020).

### **Source Selection & Justification**

The sources selected for this research primarily include peer-reviewed journal articles, government reports, and case studies published in the last five years. These sources provide valuable insights into the digital adoption challenges within the handloom sector (Brynjolfsson & McAfee, 2014; Meher et al., 2024). Reports from organizations like the **Digital Empowerment Foundation** and the **World Bank** have been prioritized, as they offer empirical evidence and comprehensive analyses of digital initiatives and their impact on rural economies (World Bank, 2020). The inclusion of these sources is justified by their relevance and credibility, allowing for a well-rounded examination of the digital barriers and opportunities facing traditional textile weavers (Sangwan & Kumar, 2024).

### **Analytical Approach**

In this study, **thematic analysis** is employed to analyze the secondary data. Key themes such as **digital illiteracy**, **infrastructure gaps**, and the **economic impact** of digital exclusion are extracted and examined to highlight the barriers faced by weavers (Singh et al., 2015; Potnis, 2016). By synthesizing the findings from various sources, the paper evaluates the effectiveness of existing digital initiatives and explores potential solutions to bridge the digital divide (Brynjolfsson & McAfee, 2014). Additionally, a **comparative analysis** is conducted to assess the success of various digital empowerment programs and their applicability to rural handloom weavers (Priya & Shrivastava, 2020). This analytical approach helps in identifying the most effective strategies for integrating digital tools and infrastructure into the traditional weaving sector.

### **Scope & Limitations**

This research is based entirely on secondary data, meaning the findings are limited to existing literature and reports. The absence of primary data collection restricts the ability to assess real-time conditions of rural weavers and may limit the depth of the findings.

Additionally, while a broad range of studies is included, the analysis is constrained by the available data, which may not capture the full diversity of weavers' experiences across different regions of India (Casciani et al., 2022). Despite these limitations, the secondary data analysis provides valuable insights into the systemic challenges and opportunities for digital inclusion in the handloom sector.

## **Findings and Analysis**

### **Barriers to Digital Adoption**

Digital exclusion remains a significant challenge for rural handloom weavers, as over 70% of weavers report low levels of digital literacy, and 60% lack reliable internet access (Sangwan & Kumar, 2024; Meher et al., 2024). These barriers hinder their ability to expand their businesses and access broader markets (Majumdar & Sinha, 2018). Moreover, limited access to smartphones and poor digital infrastructure further exacerbates the issue (Bach et al., 2013; Singh et al., 2018). The reliance on intermediaries, who control the supply chain, further limits profitability and the weavers' ability to engage directly with consumers (Digital Empowerment Foundation, 2021; Ghosal et al., 2020). Thus, these constraints undermine the economic mobility and growth potential of rural weavers, reinforcing their exclusion from the digital economy.

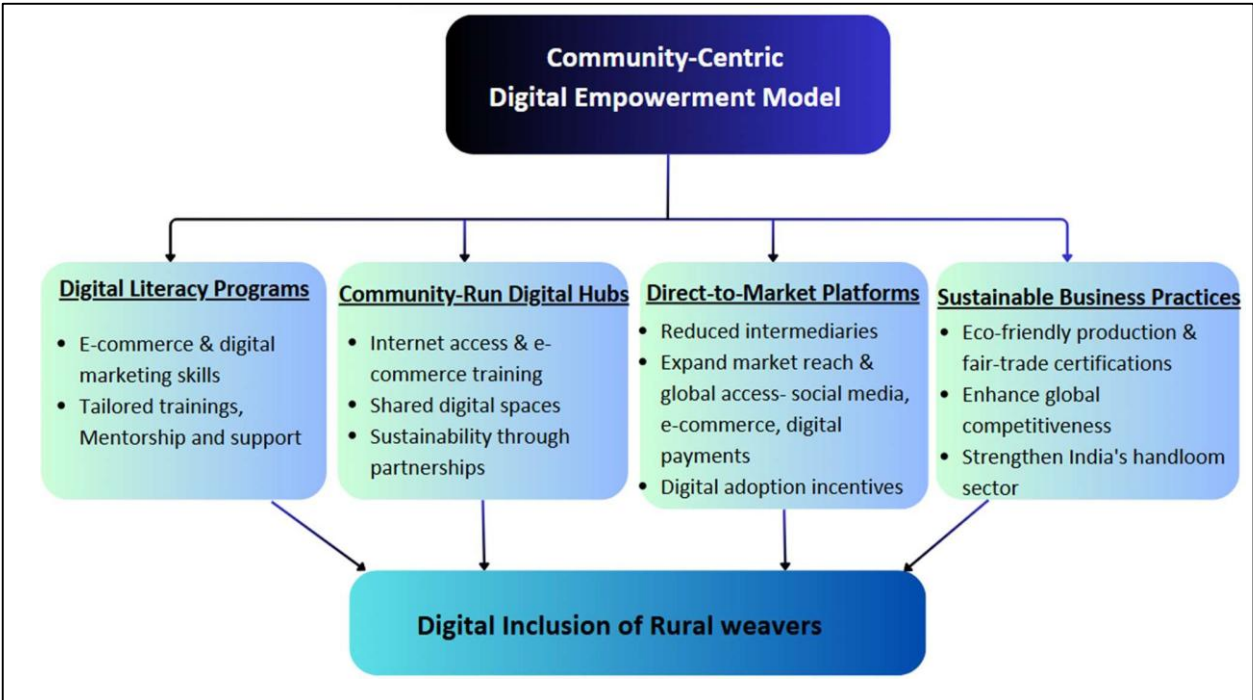
## **Discussion**

### **Proposed Empowerment Model and Policy Recommendations**

The **Community-Centric Digital Empowerment Model** as demonstrated in Figure 1 is proposed to address the digital exclusion of handloom weavers, focusing on digital literacy, infrastructure development, direct-to-market strategies, and adopting sustainable business practices.



Figure 1      *Proposed Community-Centric Digital Empowerment Model*



Firstly, Bridging the digital divide for traditional textile weavers in rural India requires targeted **digital literacy programs** tailored to their specific needs. Government initiatives, such as the 'DigiGaon' (Digital Village) program and Common Service Centers (CSCs), have played a crucial role in improving digital access (Digital Empowerment Foundation, 2021; World Bank, 2020). However, many weavers continue to face challenges due to the lack of hands-on training. To address this gap, specialized workshops focused on e-commerce platforms, digital payment systems, and social media marketing can empower weavers to expand their market reach (Meher et al., 2024; Ansari & Tripathi, 2018). For instance, the E-Santa initiative by the Indian government has successfully enabled artisans and small-scale producers to sell their goods online, showcasing the transformative potential of digital inclusion in rural economies (Sangwan & Kumar, 2024). By implementing similar programs

with localized language support and hands-on mentorship, digital literacy efforts can significantly uplift weavers and integrate them into the modern marketplace.

Second, the creation of **community-run digital hubs** equipped with internet access and e-commerce training is essential to bridging the digital infrastructure gap among rural weavers. These hubs can serve as shared spaces where artisans develop digital skills, access online marketplaces, and engage in e-commerce activities (Digital Empowerment Foundation, 2021; World Bank, 2020). To ensure the sustainability and scalability of these hubs, collaboration among local governments, NGOs, and private sector stakeholders is crucial. For example, partnerships with organizations such as the Digital Empowerment Foundation and initiatives like India's Common Service Centers (CSCs) have demonstrated the potential of these hubs to empower rural entrepreneurs by providing them with essential digital skills, access to e-commerce platforms, and opportunities for economic growth (Priya & Shrivastava, 2020). By fostering digital inclusion through these community-driven efforts, the hubs can generate long-term economic opportunities, helping weavers thrive in the digital economy.

Moreover, promoting **direct-to-market platforms** can significantly reduce the influence of intermediaries. By using digital tools such as e-commerce websites, mobile payment solutions, and social media marketing, weavers can expand their reach and tap into global markets (Sangwan & Kumar, 2024; Kumar et al., 2022). Government policies, such as the Digital India Initiative, National Handloom Development Programme (NHDP), and Mahatma Gandhi Bunkar Bima Yojana, should incentivize digital adoption by implementing financial aid programs, offering subsidies for smartphones, internet access, and training programs, as well as reducing transaction fees on digital platforms (Carpenter & Poon, 2023; Kanupriya, 2022).

Finally, integrating **sustainable business practices**, such as eco-friendly production methods like natural dyeing techniques, using organic cotton, and implementing water-efficient weaving methods, along with fair-trade certifications like Handloom Mark, which certifies that fabric is authentically handwoven and GOTS certification, which guarantees the use of organic materials and eco-friendly dyeing techniques, would not only empower rural weavers but also create a more competitive and attractive market for their products. This, in turn, can encourage wider digital adoption by increasing their access to global markets, thus helping to bridge the digital divide and strengthening India's handloom sector (Digital Empowerment Foundation, 2021; Priya & Shrivastava, 2020).

### Counterarguments

#### Preservation of Traditional Practices vs. Digital Adoption

While the **Community-Centric Digital Empowerment Model** offers significant promise, several counterarguments must be considered. One of the primary challenges is the **limited digital infrastructure** in rural areas, which impedes access to reliable internet and digital tools (Sangwan & Kumar, 2024; PLACE, 2018). Despite the availability of digital tools, the high **cost of adoption** could be prohibitive, especially for artisans with constrained financial resources. This could result in unequal access to the digital economy, disproportionately benefiting those with more economic means (Meher et al., 2024). Furthermore, establishing the **infrastructure** necessary for digital hubs and training programs requires substantial financial investment, and the **implementation process** may be slow due to **budget constraints** in rural regions (Priya & Shrivastava, 2020).

Additionally, **cultural resistance** poses a considerable obstacle. Many weavers are hesitant to adopt digital tools due to **lack of understanding** or **trust in technology** (Ansari & Tripathi, 2018). This resistance is rooted in concerns that digital adoption could undermine

traditional practices or alter the cultural value embedded in their craft (Kumar et al., 2022).

Addressing these concerns requires **concerted efforts** to build trust in digital technologies and ensure that these tools enhance, rather than replace, traditional methods (Ibragimova, 2019).

Despite these challenges, the proposed model can be strengthened through **collaborative efforts** from stakeholders in the private, public, and non-profit sectors. Investments in infrastructure, coupled with education campaigns focused on the benefits of digital tools, can mitigate cultural resistance and ensure that digital adoption complements traditional practices, rather than diminishing them (Priya & Shrivastava, 2020; Meher et al., 2024).

## Conclusion

### Towards a Community-Centric Digital Empowerment Model

This study examines the digital exclusion faced by traditional handloom weavers in rural India and introduces the **Community-Centric Digital Empowerment Model** to bridge this gap. The research emphasizes the transformative potential of digital empowerment to enhance market access, economic sustainability, and business growth for rural artisans (Sangwan & Kumar, 2024; Priya & Shrivastava, 2020; Mishra & Mohapatra, 2020). The key findings suggest that addressing **digital literacy, infrastructure limitations, and financial barriers** is crucial for enabling equitable digital adoption and empowering weavers to engage in the digital economy (Meher et al., 2024; Kumar et al., 2022).

This study contributes to the broader discourse on **digital inclusion** by highlighting the importance of integrating technology into traditional sectors without compromising cultural values (Digital Empowerment Foundation, 2021). The proposed model aims to provide weavers with the skills, tools, and support needed to expand their market reach and **improve profitability**, ensuring they are not excluded from the digital age (Sangwan & Kumar, 2024).

However, limitations include the **lack of longitudinal studies** to assess the long-term impacts of digital empowerment on rural communities (Priya & Shrivastava, 2020). Future research should focus on **evaluating** the effectiveness of these strategies and exploring their broader impact on rural economies (Kumar et al., 2022; Yadav et al., 2020).

### **Implications and Future Research**

The implications of this research extend beyond the handloom sector, contributing to broader discussions on **digital transformation** and its role in enhancing **economic inclusivity**. By facilitating **direct market access** for rural artisans, digital tools can reduce dependency on intermediaries, thereby increasing profitability (Brynjolfsson & McAfee, 2014; Meher et al., 2024). To improve these outcomes, future research should explore the **long-term sustainability** of digital initiatives and evaluate their effectiveness in other rural sectors (Singh et al., 2018; Ta & Lin, 2023). Additionally, further studies should assess the **scalability** of the proposed model across diverse geographical and cultural contexts to determine its **universal applicability** (Digital Empowerment Foundation, 2021).

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